

1.

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C++ PROGRAMMING

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192421129RC

QUESTIONS 9 / 9 completed

Area

AREA

Write a function to calculate the area of a circle given its radius.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <iomanip>
3 using namespace std;
4
5 // Function to calculate the area of a circle
6 double calculateArea(double radius) {
7     const double PI = 3.141592653589793;
8     return PI * radius * radius;
9 }
10
11 int main() {
12     double radius;
13     cin >> radius;
14
15     cout << fixed << setprecision(2) << calculateArea(radius) << endl;
16
17     return 0;
18 }
```

INPUT

5

2.

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QUESTIONS 9 / 9 completed

Convert

CONVERT

Write a function to convert a given temperature from Celsius to Fahrenheit.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 #include <iomanip>
3 using namespace std;
4
5 // Function to convert Celsius to Fahrenheit
6 double celsiusToFahrenheit(double celsius) {
7     return (celsius * 9.0 / 5.0) + 32.0;
8 }
9
10 int main() {
11     double celsius;
12     cin >> celsius;
13
14     cout << fixed << setprecision(2) << celsiusToFahrenheit(celsius) << endl;
15
16     return 0;
17 }
```

INPUT

25

3.



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QUESTIONS 3 / 20 completed

Bank Account Class: ▾

BANK ACCOUNT CLASS

Bank Account Class:
Create a C++ class representing a bank account with functionalities such as deposit, withdrawal, and balance inquiry.
Sample Input:
Account Number:
1234567890
Account Holder Name:
John Doe
Balance: 5000

PROGRAM EDITOR C++ ▾

Run Save

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class BankAccount {
6 private:
7     string accountNumber;
8     string accountHolderName;
9     double balance;
10
11 public:
12     // Constructor to initialize account details
13     BankAccount(string accNum, string name, double initi
14         accountNumber = accNum;
15         accountHolderName = name;
16         balance = initialBalance;
17     }
18
19     // Function to deposit money
20     void deposit(double amount) {
21
```

INPUT

1234567890
John Doe
5000
2000
1000

simatslab.saveetha.com/practice_app/dashboard_student

4.

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QUESTIONS 3 / 20 completed

Student Record Sys ▾

STUDENT RECORD SYSTEM

Student Record System: Design a class to manage student records including data such as name, ID, and grades. Implement functions to calculate the average grade and display student information. Sample Input: Student Name: Jane Doe

PROGRAM EDITOR C++ Run Save

```
1 #include <iostream>
2 #include <string>
3 #include <vector>
4 #include <numeric>
5 #include <iomanip>
6
7 using namespace std;
8
9 class Student {
10 private:
11     string name;
12     int id;
13     vector<double> grades;
14
15 public:
16     // Function to set student details
17     void setDetails(string studentName, int studentID, v
18         name = studentName;
19         id = studentID;
20         grades = studentGrades;
21
```

INPUT

Jane Doe
12345
85.5 90 87 92.5

5.

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QUESTIONS 3 / 20 completed

Employee Manager ▾

EMPLOYEE MANAGEMENT

Employee Management: Develop a class for an employee with attributes like name, salary, and employee ID. Include methods to raise the salary and display employee details. Sample Input: Employee Name: John Doe Employee ID: E12345 Salary: 10000

PROGRAM EDITOR C++ Run Save

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Employee {
6 private:
7     string name;
8     string id;
9     double salary;
10
11 public:
12     // Function to set employee details
13     void setDetails(string empName, string empID, double
14         name = empName;
15         id = empID;
16         salary = empSalary;
17     }
18
19     // Function to raise salary
20     void raiseSalary(double amount) {
21
```

INPUT

John Doe
E12345
10000
5000